PTVA's

MULUND COLLEGE OF COMMERCE (AUTONOMOUS)

Programme: B.Sc. Information Technology **Class: TY** **Semester: V**

Exam: ESE **Name of the Course: Big Data and NOSQL**

Date of Exam: 07/10/2025 **Marks: 60** **Time: 2 hours**

Instructions:

(1) **All** questions are **compulsory**.
(2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
(3) Answers to the **same question** must be **written together**.
(4) Numbers to the **right** indicate **marks**.
(5) Draw **neat labeled diagrams** wherever **necessary**.
(6) Use of non-programmable calculator is **allowed**.

Q 1 Attempt *any TWO* of the following: **12**

A. Explain the concept of Big Data in your own words. Identify and describe various sources from which Big Data is generated.

B. Define the CAP theorem and list its three components.

C. How would you apply MongoDB's design philosophy when developing a scalable database solution for a social media platform?

D. Analyze the features of MongoDB and explain how they are different from those of a relational database.

Q 2 Attempt *any TWO* of the following: **12**

A. Write the MongoDB command to create the following with an example.
(i) Database; (ii) Collection; (iii) Document; (iv) Drop Collection; (v) Drop Database; (vi) Index.

B. In what scenarios is capped collection useful in MongoDB?

C. List the three core components of the MongoDB package and briefly describe their roles.

D. Analyze the structure and purpose of Binary JSON (BSON) in MongoDB.

Q 3 Attempt *any TWO* of the following: **12**

A. List the key limitations of sharding in MongoDB and briefly describe them.

B. What is Data Storage engine? Which is the default storage engine in MongoDB? Also compare MMAP and Wired Tiger storage engines.

C. Write a short note on the replication lag in MongoDB and mention the situations in which it may occur.

D. Describe the purpose of journaling in MongoDB with the help of a neat diagram.

Q 4 Attempt *any TWO* of the following: **12**

A. Describe the architecture of TimesTen with the help of a neat diagram.

B. Evaluate the concept of Ajax and its uses in web development. Assess how Ajax can be implemented with jQuery and justify its effectiveness in enhancing web application interactivity.

C. Demonstrate how to add and remove elements from the DOM using jQuery with suitable example.

D. What is a Solid-State Drive (SSD)? List its key characteristics.

Q 5 Attempt *any TWO* of the following: **12**

A. Explain the different data types supported by JSON with an example.

B. Analyze and compare JSON and XML in terms of structure, readability and typical use cases.

C. List and describe the components of an HTTP request message.

D. Write a short note on persisting JSON data and how it can be saved with an example.